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HTV

HALF YEARLY EXAMINATION - 2022

XII - Std

MATHEMATICS

Time : 3.00 Hrs

Marks : 90

Section - A

I **Choose the correct answer :-** 20 X 1 = 20

1. If $A=\begin{bmatrix}2 & 3\\ 5 & -2\end{bmatrix}$ be such that $\lambda A^{-1}=A$ then λ is.

   a) 17 b) 14 c) 19 d) 21

2. If $P=\begin{bmatrix}1 & x & 0\\ 1 & 3 & 0\\ 2 & 4 & -2\end{bmatrix}$ is a adjoint of 3 X 3 matrix A and $|A|=4$, then x is.

   a) 15 b) 12 c) 14 d) 11

3. The value of $i^{2018}+i^{2019}+i^{2020}+i^{2021}$ is

   a) 0 b) 1 c) –1 d) i

4. The solution of $|z|-z=1+2i$ is.

   a) $\frac{3}{2}-2i$ b) $\frac{-3}{2}-2i$ c) $2-\frac{3}{2}i$ d) $2+\frac{3}{2}i$

5. A zero of x^3+64 is.

   a) 0 b) 4 c) 4i d) –4

6. $\sin^{-1}\left(\frac{x}{5}\right)+\cos ec^{-1}\left(\frac{5}{4}\right)=\frac{\pi}{2}$ then x is

   a) 4 b) 5 c) 3 d) 2

7. If the function $f(x)=\sin^{-1}(x^2-3)$, then x belong to

   a) [–1, 1] b) $\left[\sqrt{2},2\right]$ c) $\left[-2,-\sqrt{2}\right]\cap\left[\sqrt{2},2\right]$ d) $\left[-2,-\sqrt{2}\right]\cup\left[\sqrt{2},2\right]$

8. If $x+y=k$ is a normal to the parabola $y^2=12x$ then the value of k is

   a) 3 b) –1 c) 9 d) 1

9. The distance betweenthe foci of the ellipse $9x^2+5y^2=180$ is

   a) 2 b) 4 c) 6 d) 8

10. If $\left[\vec{a}X\vec{b}\ \ \vec{b}X\vec{c}\ \ \vec{c}X\vec{a}\right]=121$ then $\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]=$ is

    a) $\frac{121}{2}$ b) 8 c) 11 d) 121

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11. If $\vec{a}=-\vec{i}+\vec{j}+\vec{k},\ \vec{b}=\vec{i}+\vec{j},\ \vec{c}=\vec{j}$ and $(\vec{a}\ X\ \vec{b})\ X\ \vec{c}=\lambda\vec{a}+\mu\vec{b}$ then $\lambda+\mu$ is

a) 0 b) 1 c) 2 d) 3

12. The minimum value of the function $|3-x|+9$ is

a) 0 b) 3 c) 6 d) 9

13. Angle between $y^2=x$ and $x^2=y$ at origin is

a) $\frac{\pi}{2}$ b) $\frac{\pi}{3}$ c) $\frac{\pi}{6}$ d) can't find

14. $V(x, y)=\log(e^x+e^y)$ then $\frac{\partial V}{\partial x}+\frac{\partial V}{\partial y}=$

a) e^x+e^y b) $\frac{1}{e^x+e^y}$ c) 2 d) 1

15. The value of $\int_{-4}^{4}|x+3|\,dx$ is

a) 50 b) 25 c) 16 d) 8

16. The value of $\int_{0}^{\pi}\sin^4 x\ dx$ is

a) $\frac{3\pi}{10}$ b) $\frac{3\pi}{8}$ c) $\frac{3\pi}{4}$ d) $\frac{3\pi}{2}$

17. The general solution of the differential equation $\frac{dy}{dx}=\frac{y}{x}$ is

a) $xy+k$ b) $y=k\log x$ c) $y=kx$ d) $\log y=kx$

18. The solution of $\frac{dy}{dx}=2^{y-x}$ is

a) $2^x+2^y=c$ b) $2^x-2^y=c$ c) $\frac{1}{2^x}-\frac{1}{2^y}=c$ d) $x+y=c$

19. If $U=y\sin x$ then $\frac{\partial^2 y}{\partial x\partial y}=$

a) $\cos x$ b) $\cos y$ c) $\sin x$ d) $\sin y$

20. If $\frac{z-1}{z+1}$ is purely imaginary then $|z|$ is

a) $\frac{1}{2}$ b) 1 c) 2 d) 3

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Section - B

II **Answer any 7 questions. Q.No. 30 is compulsory :-** 7X2=14

21. If $adjA = \begin{pmatrix} 0 & -2 & 0 \\ 6 & 2 & -6 \\ -3 & 0 & 6 \end{pmatrix}$ then find A^{-1}.

22. Show that $\left(2+i\sqrt{3}\right)^{10} - \left(2-i\sqrt{3}\right)^{10}$ is purely imaginary.

23. Solve : $x^4 - 14x^2 + 45 = 0$.

24. Find the value of $\cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}(-1)$.

25. Obtain the equation of the circle for which (–4, –2) and (1,1) are the ends of a diameter..

26. Verify that the vectors $\vec{i} + 2\vec{j} - 3\vec{k}$, $2\vec{i} - \vec{j} + 2\vec{k}$, $3\vec{i} + \vec{j} - \vec{k}$ are coplanar.

27. Solve : $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$

28. Evaluate : $\int_0^1 x^4(1-x)^5\,dx$

29. If V (x, y, z) = xy + yz + zx then find dv.

30. Prove that the function $f(x) = x^2 - 2x - 3$ is strictly increasing in $(2,\infty)$.

Section - C

Answer any 7 questions. Q.No. 40 is compulsory :- 7 X 3 = 21

31. Evaluate. $\lim_{x \to 1} x^{\frac{1}{1-x}}$

32. U (x, y, z) = log $(x^3 + y^3 + z^3)$ then find $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$.

33. Solve : $\frac{dy}{dx} + \frac{y}{x} = \sin x$

34. Show that the shortest distance between the lines

$\vec{r} = (6\vec{i} + \vec{j} + 2\vec{k}) + S(\vec{i} + 2\vec{j} - 3\vec{k})$

$\vec{r} = (3\vec{i} + 2\vec{j} - 2\vec{k}) + t(2\vec{i} + 4\vec{j} - 5\vec{k})$ and is $\frac{7\sqrt{5}}{5}$.

35. If the normal at thepoint 't_1' on the parabola $y^2 + 4ax$ meets the parabola again at the point 't_2' then prove that $t_2 = -\left(t_1 + \frac{2}{t_1}\right)$.

36. Solve : $\tan^{-1}(2x) + \tan^{-1}(3x) = \frac{\pi}{4}$, $6x^2 < 1$.

37. If P is real, discuss the nature of the roots of the equation $4x^2 + 4Px + P + 2 = 0$, intermsof P.

38. Show that the equation $z^2 = \bar{z}$ has foursolutions.

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39. Find the rank of the matrix $\begin{bmatrix} 1 & 1 & 1 & 3 \\ 2 & -1 & 3 & 4 \\ 5 & -1 & 7 & 11 \end{bmatrix}$.

40. Evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{dx}{1+\sqrt{\tan x}}$

Section - IV

Answer all questions :- 7 X 5 = 35

41. a) Investigate for what values of λ, μ the system of linear equations. $x+2y+z=7;\ x+y+\lambda z=\mu;\ x+3y-5z=5$ has i) no solution ii) a unique solution iii) an infinite number of solution. ***(OR)***

b) If $z = x + iy$ and $\arg\left(\frac{z-1}{z+1}\right)=\frac{\pi}{2}$, show that $x^2 + y^2 = 1$.

42. a) Solve : $x^4 - 10x^3 + 26x^2 - 10x + 1 = 0$. ***(OR)***

b) Find the domain of $f(x)=\sin^{-1}\left(\frac{|x|-2}{3}\right)+\cos^{-1}\left(\frac{1-|x|}{4}\right)$.

43. a) A bridge has a parabola arch that is 10m high in the centre and 30m wide at the bottom. Find the height of the arch 6m from the centre, on either sides. ***(OR)***

b) Prove by vector method $\sin(\alpha+\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta$.

44. a) Sketch the curve $y = f(x) = x^2 - x - 6$. ***(OR)***

b) If $u=\tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, show that $x\frac{\partial u}{\partial x}+y\frac{\partial u}{\partial y}=\sin 2u$.

45. a) Find the area of the region bounded between the curves $y = \sin x$ and $y = \cos x$ and the lines $x = 0$ and $x = \pi$. ***(OR)***

b) A pot of boiling water at 100°C is removed fronm a stove at time t = 0 and left to cool in the kitchen. After 5 minutes, the water temperature has decreased to 80°C and another 5minutes later it has dropped to 65°C. Determine thetemperature of the kitchen.

46. a) Find the volume of a right - circular cone of base radius 'r' and height 'h'. ***(OR)***

b) Find the parametrix vector and Cartesian equation of the planecontaining the line $\frac{x-2}{2}=\frac{y-2}{3}=\frac{z-1}{-2}$ and passing through the point $(-1, 1, -1)$.

47. a) If the curves $ax^2 + by^2 = 1$ and $cx^2 + dy^2 = 1$ intersect each other orthogonally then show that $\frac{1}{a}-\frac{1}{b}=\frac{1}{c}-\frac{1}{d}$. ***(OR)***

b) Find the equation of the circle passing through the points (1, 1) (2, –1) and (3, 2).

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